



La₂O₃-tailored thermal compatible and stable glass-ceramics seals for solid oxide electrolysis cells

Zihao Xu ^{a b c}, Fushun Zhang ^{a b} , Xu Dang ^d, Chaoyi Wang ^{a b}, Xiaobo He ^e , Hongtao Chang ^{a b}, Jun Peng ^{a b},
Liuzhen Bian ^{a c} , Yunting Hou ^e, Shengli An ^{a b c}

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Abstract

The development of durable sealing materials remains a critical challenge for solid oxide electrolysis cells (SOECs) technology despite its great promise in green hydrogen production. Unlike solid oxide fuel cells (SOFCs) sealing systems, SOECs sealants must withstand extreme dual-atmosphere conditions (simultaneous oxidizing/wet-reducing environments) during prolonged high-temperature steam electrolysis. To bridge this technological gap, we present a lanthanum oxide-engineered diopside glass-ceramic (La₂O₃-CMASBZ) through systematic composition-structure optimization. The incorporation of La₂O₃ induces significant structural modifications within the glass network. These changes manifest a drop in the glass transition temperature (T_g) from 654.4°C to 632.4°C and enhanced thermal compatibility, evidenced by a progressive rise in the coefficient of thermal expansion

(CTE) from 9.58 to $13.7 \times 10^{-6} \text{K}^{-1}$. Crucially, the optimized sealant exhibits exceptional interfacial stability during rigorous 100-h dual-atmosphere aging tests, maintaining structural integrity at critical interfaces with yttria-stabilized zirconia (YSZ) electrolyte, Ni-YSZ hydrogen electrode, and SUS430 stainless steel interconnector. This work demonstrates that La_2O_3 simultaneously tailors glass network dynamics and enhances electrochemical stability and durability, providing a materials platform for next-generation SOECs.



Keywords

Solid oxide electrolysis cells; Sealing; Glass-ceramic; Coefficient of thermal expansion

1. Introduction

The global transition toward carbon-neutral energy systems has positioned hydrogen energy as a strategic priority due to its exceptional energy density and carbon-free combustion characteristics [[1], [2], [3]]. However, approximately 95% of current hydrogen production still relies on steam methane reforming (SMR), a process that generates 9–12 kg $\text{CO}_2/\text{kg H}_2$ [4], fundamentally conflicting with decarbonization objectives. Proton exchange membrane water electrolysis (PEMWE) and solid oxide electrolysis cells (SOECs) have emerged as promising alternatives for green hydrogen production when powered by renewable electricity [5]. Notably, SOECs technology demonstrates superior thermodynamic advantages through high-temperature operation (600–800 °C), achieving electrical-to-hydrogen conversion efficiencies exceeding 90% [6]. This represents a 30% improvement over conventional alkaline electrolyzers [7,8] attributable to reduced ohmic losses and enhanced reaction kinetics at elevated temperatures. Furthermore, the SOECs technology enables effective utilization of industrial waste heat via thermal integration systems [[9], [10], [11], [12]].

Despite SOECs exhibiting compelling performance metrics, their industrial deployment faces critical material challenges, especially in sealing reliability. The sealing components must maintain hermeticity under extreme operational conditions

(H₂O/H₂ and O₂). The CTE of sealing material must align with neighbouring components during long-term operation (YSZ electrolyte: $10.5 \times 10^{-6} \text{K}^{-1}$; Crofer 22 APU interconnection: $12.5 \times 10^{-6} \text{K}^{-1}$) [[13], [14], [15]]. Current sealing technologies, predominantly glass-ceramic composites, suffer from three primary failure modes: (1) Interfacial delamination caused by CTE mismatch [16]; (2) Chemical degradation under humid reducing atmospheres [[17], [18], [19]]; (3) Mechanical fatigue during thermal cycling [[20], [21], [22]]. These failures induce gas crossover, efficiency decay, and catastrophic stack failure, ultimately compromising system durability [21,22]. A critical failure pathway involves steam-induced corrosion of glass matrices. Above 700°C, water vapor reacts with modifier cations (e.g., Na⁺, Ca²⁺) in silicate glasses through ion-exchange reactions, leading to network depolymerization through hydrolysis of bridging oxygen bonds. This process results in volatile hydroxyl species formation, structural reorganization, and surface devitrification accompanied by porosity development [23], severely degrading both hermeticity and mechanical strength [[24], [25], [26], [27]].

Calcium-magnesium-alumina-silicate (CAMS) glass-ceramics containing diopside (CaMgSi₂O₆) and anorthite (CaAl₂Si₂O₈) crystalline phases demonstrate improved steam resistance by suppressing alkali metal mobility. However, conventional CAMS systems face two intrinsic limitations: (1) elevated softening temperatures (>850°C) that impede effective sealing formation, and (2) inherent microporosity (>5%) that accelerates stress concentration under thermal cycling [[28], [29], [30], [31]]. Recent advances in rare-earth doping strategies reveal promising modifications. Luo et al. demonstrated through Fourier-transform infrared spectroscopy (FTIR) that 2.5 mol% La₂O₃ incorporation increases the non-bridging oxygen index (NBO/T) from 1.8 to 2.3, concomitant with crystallization of Ca₃La₆(SiO₄)₆ phases that enhance CTE from $8.2 \times 10^{-6} \text{K}^{-1}$ to $9.5 \times 10^{-6} \text{K}^{-1}$ [32]. Although the Ca_{0.7}Sr_{0.3}Mg_{0.8}Al_{0.1}La_{0.1}Si_{1.9}O₆ system achieves reduced softening temperature (795°C) and low porosity (<0.8%), its performance remains unvalidated under actual SOECs operating conditions [33].

In this study, we report a compositionally engineered CaO-MgO-SiO₂-Al₂O₃-B₂O₃-ZrO₂-La₂O₃ (CMASBZ-L) glass-ceramic system with controlled La₂O₃ doping (0–3.0 wt%). Systematic composition optimization reveals that La₂O₃ incorporation induces three critical improvements: (1) reduced glass transition temperature (T_g) to 715°C and softening point (T_s) to 780°C, facilitating in-situ sealing formation; (2) tailored CTE values spanning $9.58\text{--}13.7 \times 10^{-6} \text{K}^{-1}$, achieving near-perfect thermal compatibility with SOECs stack components (YSZ electrolyte: $10.5 \times 10^{-6} \text{K}^{-1}$; metallic interconnects: $12.5 \times 10^{-6} \text{K}^{-1}$); and (3) enhanced hydrothermal stability under operational extremes. The optimized 1.0 wt% La₂O₃-doped CMASBZ exhibited remarkable electrochemical performance, demonstrating an open-circuit voltage (OCV) of 1.01 V under humidified hydrogen—equivalent to 82% retention of the theoretical Nernst potential (1.23 V at 800°C).

2. Experimental section

2.1. Material preparation

The glass-ceramic samples were synthesized using high-purity raw materials: CaCO_3 (99.9%), B_2O_3 (99.9%), MgO (99.9%), Al_2O_3 (99.9%), SiO_2 (99.9%), ZrO_2 (99.9%), and La_2O_3 (99.9%). The raw materials ($27.25\text{CaO}-34.32\text{MgO}-1.36\text{Al}_2\text{O}_3-12.73\text{SiO}_2-23.41\text{B}_2\text{O}_3-0.93\text{ZrO}_2$) with varying $x\text{La}_2\text{O}_3$ ($x=0.0, 0.5, 1.0, 3.0\text{wt}\%$) were weighed, mixed uniformly in a polyethylene canister, transferred to a graphite crucible, and heated at 1550°C for 2h in a muffle furnace. Post-melting, part was cast into a square mold, while the remainder was quenched in deionized water. The resulting glass frit was ball-milled and sieved through a 200-mesh sieve to obtain the final precursor powder. The Ni-YSZ hydrogen electrode was prepared using a slurry of 60wt% NiO and 40wt% 8YSZ (ZrO_2 stabilized with 8% mol Y_2O_3). The SUS430 stainless steel was selected as an interconnect material. The resultant glass-ceramic specimens were designated as $x\text{La}_2\text{O}_3\text{-CMASBZ}$ ($x=0.0, 0.5, 1.0, 3.0$).

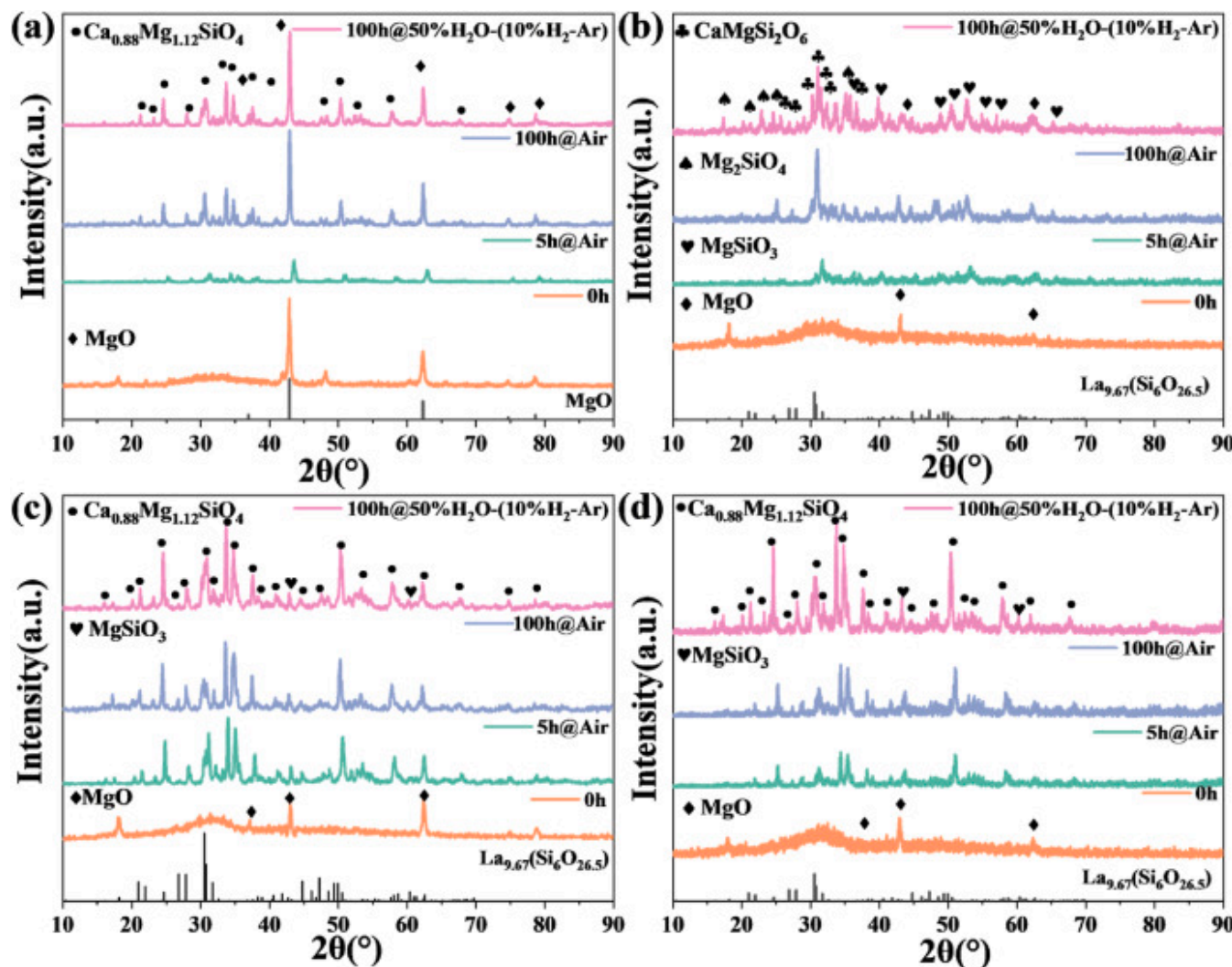
2.2. Characterization

The phase structures were analysed by an X-ray diffractometer (MiniFlex 600, Rigaku Corporation, Japan) equipped with Cu-K α radiation ($\lambda=0.15418\text{nm}$) at a step scanning rate of $5^\circ/\text{min}$. Raman spectra were collected using a Renishaw spectrometer (532nm laser wavelength). The differential scanning calorimetry (DSC) was conducted at heating rates of $5\text{--}20\text{Kmin}^{-1}$ under an air atmosphere on thermal analysis (STA1600, Netzsch, Germany). The thermal expansion behaviour was measured by a dilatometer (402 Expedit Supreme, Netzsch, Germany) at 5Kmin^{-1} up to 850°C . The dual-atmosphere aging tests (air/50% H_2O -10% H_2 -Ar) were conducted on a SUS430/glass-ceramic/Ni-YSZ/YSZ/glass-ceramic stack to simulate SOECs operation. Post-test samples were sectioned, polished, and analysed by scanning electron microscopy (SIGMA-300, Zeiss Merlin Compact, Germany). Electrochemical cells were screen-printed with NiO-YSZ/YSZ/Ag layers, and OCV was monitored under humidified hydrogen (50% relative humidity at 80.4°C).

3. Results and discussion

3.1. Phase stability analysis

Glass-ceramics exhibit characteristics intermediate between glasses and ceramics. Typically, these materials are amorphous before heat treatment, while achieving varying degrees of crystallization through modifications in their glass composition after heat treatment. Fig. 1(a) shows the XRD patterns of four kinds of glass-ceramics with various La₂O₃ content under different treatment conditions. The MgO peaks (at ~36.8°, ~42.8°, and 62.2°) and an amorphous peak at about 30° are observed in the xLa₂O₃-CMASBZ without heat treatment. The formation of MgO is mainly due to its high melting point (~2852°C), small ionic radius, and fast migration rate, which leads to rapid crystallization during the rapid cooling process. Fig. 1(b–d) demonstrate that La₂O₃-doped glass suppresses phase separation, evidenced by reduced MgO peak intensity. [34,35]. After heat treatment at 850°C, the peak at 42.8° transforms clinoenstatite (MgSiO₃) due to the reaction between MgO and SiO₂. Meanwhile, a new Lanthanum silicon oxide (La_{9.67}(Si₆O_{26.5})) (PDF#97-016-1475) precipitates in the CMASBZ glass containing La₂O₃. The primary crystalline phase shifts from synthetic (Ca_{0.88}Mg_{1.22}SiO₄) to diopside (CaMg(Si₂O₆)), forsterite (Mg₂SiO₄), clinoenstatite (MgSiO₃), then reverts back to synthetic (Ca_{0.88}Mg_{1.22}SiO₄). Further studies are required to elucidate the crystallization mechanism. Additionally, prolonged heat treatment times and the presence of reducing atmospheres (50% H₂O-(10% H₂-Ar)) enhance the intensity of the XRD diffraction peaks, consistent with previous literature reports [18,36]. Notably, no new phases emerged during these processes, suggesting that xLa₂O₃-CMASBZ (x=0, 0.5, 1.0, 3.0) possess excellent long-term stability under SOCs conditions. This stability ensures minimal variations in the thermal expansion coefficient (CTE) during operation, thereby maintaining reliable sealing performance.



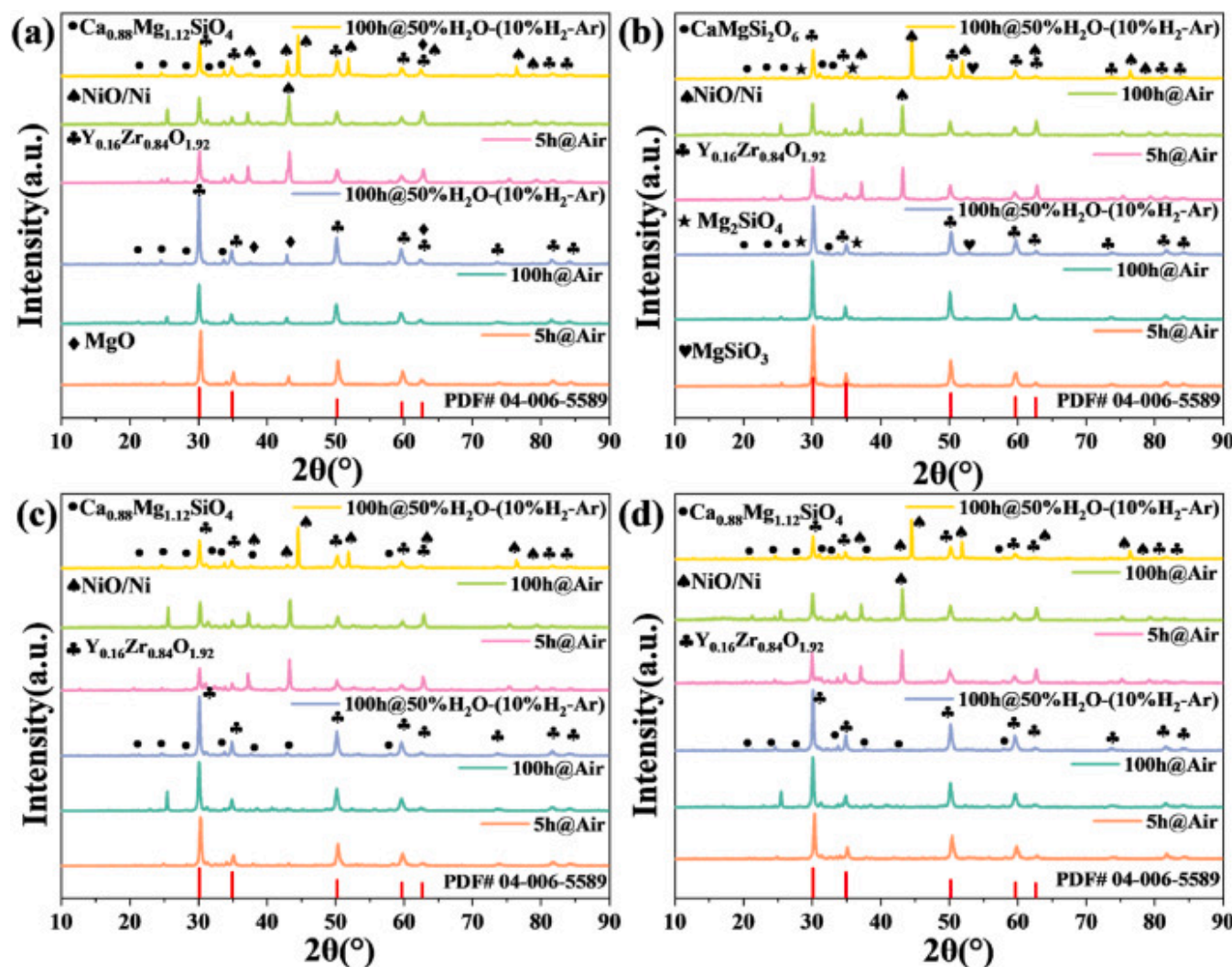
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Fig. 1. XRD patterns of the fresh glass-ceramics powder and after heating treatment under different atmospheres for 0, 5, and 100h at 850 °C. (a) 0.0 La₂O₃-CMASBZ; (b) 0.5 La₂O₃-CMASBZ; (c) 1.0 La₂O₃-CMASBZ; (d) 3.0 La₂O₃-CMASBZ.

Fig. 2 illustrates the XRD patterns of a mixture of xLa₂O₃-CMASBZ with YSZ or NiO/YSZ at a 1:1 mass ratio. Given the dominance of the diffraction peaks from YSZ and NiO/YSZ, the phase analysis primarily focuses on detecting potential crystalline phases to

evaluate whether xLa₂O₃-CMASBZ reacts with the electrolyte or NiO during processing. The xLa₂O₃-CMASBZ system (x=0.0–3.0) maintains unchanged phases with YSZ after prolonged air or reducing (50% H₂O-(10% H₂-Ar)) thermal aging, demonstrating excellent phase stability and chemical compatibility. Under reducing conditions, partial reduction of NiO to metallic Ni occurs, evidenced by attenuated NiO peaks ($\sim 42.8^\circ$, $\sim 62.9^\circ$, $\sim 75.5^\circ$, $\sim 79.5^\circ$) and emergent Ni reflections ($\sim 44.5^\circ$, $\sim 51.8^\circ$, $\sim 76.38^\circ$). Thus, the xLa₂O₃-CMASBZ glass-ceramics maintain excellent phase stability and chemical compatibility with both anode and electrolyte components during extended dual-atmosphere exposure.



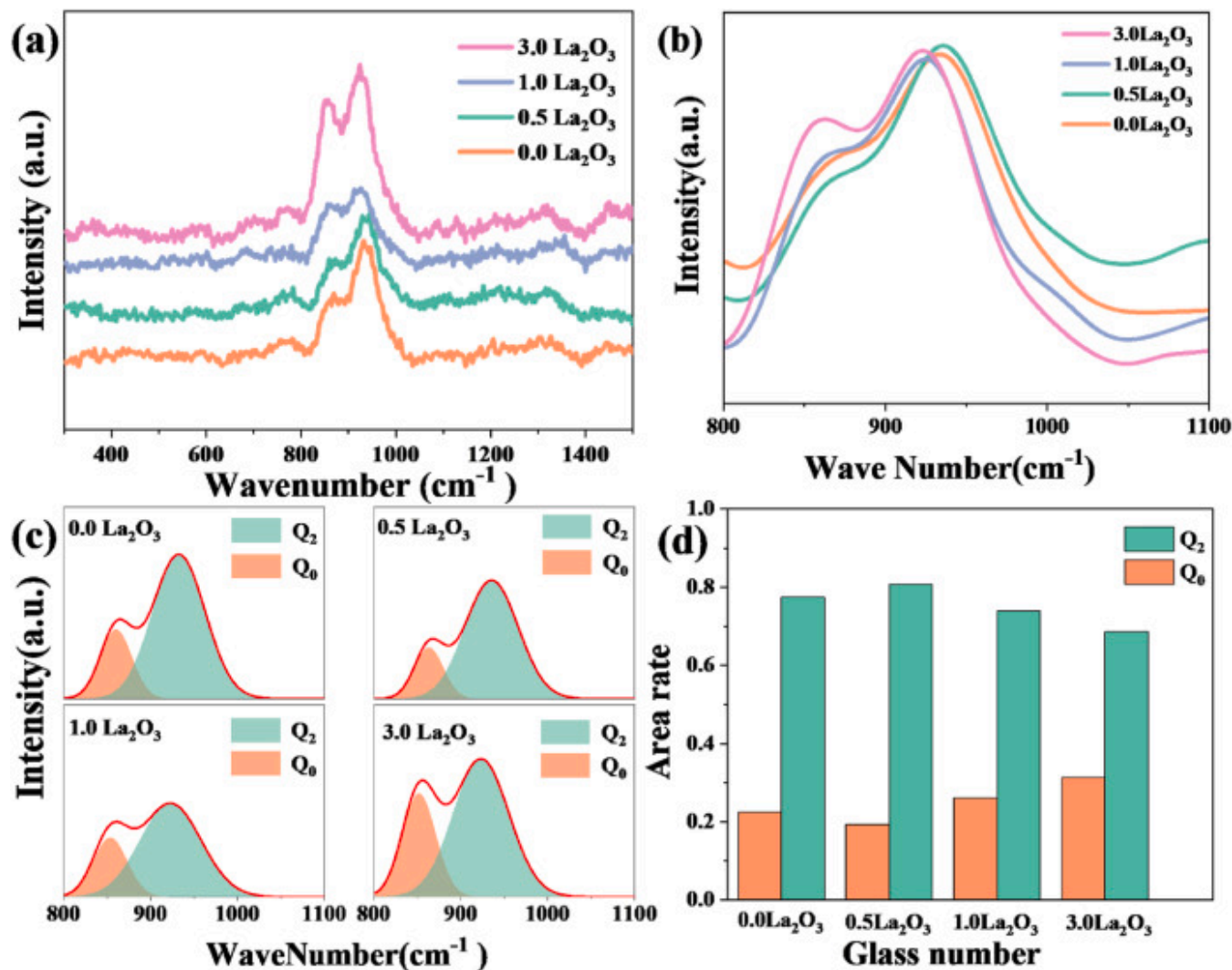
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Fig. 2. The XRD patterns of glass-ceramics with YSZ and NiO/YSZ after heating treatment at 850°C under different atmospheres for 0, 5, and 100h. (a) 0.0 La₂O₃-CMASBZ; (b) 0.5 La₂O₃-CMASBZ; (c) 1.0 La₂O₃-CMASBZ; (d) 3.0 La₂O₃-CMASBZ.

3.2. Raman analysis

The structural modification effect of La₂O₃ content on glass-ceramic was further investigated using Raman spectroscopy. Fig. 3(a and b) presents the spectra and partially enlarged detail of xLa₂O₃-CMASBZ. Three distinct regions are observed at 1200–1500cm^{−1}, 800–1200cm^{−1}, and 400–800cm^{−1}, respectively. In the low-frequency range (<400cm^{−1}), vibrational peaks primarily arise from complex distortion and torsion within the silicate network, coupled with vibrational modes of modifier cations. In the 400–800cm^{−1} region, characteristic vibrational modes of both B₂O₃ and SiO₂ structural units are identified [37]. The bending vibration of Si–O–Si bonds appears at ~430cm^{−1} and ~770cm^{−1}, while B–O–B bending and stretching vibrations are observed at ~530cm^{−1} and ~580cm^{−1}, respectively [38]. Within the high-wavenumber region (1300–1500cm^{−1}), detected spectral features originate from B₂O₃-related structural units [39,40]. The Raman spectral features in the 800–1200cm^{−1} region predominantly originate from [SiO₄] structural units. The Raman peaks initially shift to higher wavenumbers with enhanced intensity, followed by a subsequent shift to lower wavenumbers accompanied by reduced intensity and narrowed peak widths. The fitted peaks are obtained by deconvolution of the normalized data by a Gaussian function, as shown in Fig. 3(c), and the percentage of the regions occupied by the fitted units is presented in Fig. 3(d). The sharp peak at ~850cm^{−1} is assigned to Q⁰ Protosilicate (SiO₄^{4−}) units, characterized by four non-bridging oxygen (NBO) bonds per tetrahedron [[41], [42], [43]]. The sharp peak at ~950cm^{−1} corresponds to Q₂ Metasilicate (SiO₃^{2−}) units, featuring two bridging oxygen bonds and two NBOs per tetrahedron [44]. It was reported that doping with La³⁺ causes structure depolymerization and loosening [45]. The introduction of 0.5wt% La₂O₃ disrupts the [SiO₄] network, generating substantial free oxygen ions and enabling modifier cation integration. The area about Q₂ of 0.5La₂O₃-CMASBZ increases rapidly, which disrupts the [SiO₄] network, generating substantial free oxygen ions and enabling modifier cation integration. At higher La₂O₃ concentrations, intensified network fragmentation drives Q₂-to-Q₀ reconversion, accompanied by increased free oxygen content. This structural disordering manifests as reduced network rigidity and viscosity, directly correlating with decreased glass transition (T_g) and softening (T_s) temperatures. The addition of a small amount of La₂O₃ increases the coefficient of thermal expansion (CTE) by disrupting the glass network, whereas excessive La³⁺ ions may reduce the CTE through an accumulative effect on the glass structure [32].



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Fig. 3. (a) Raman spectra of four as-quenched sample with different La_2O_3 content; (b) Normalized results of Raman spectral data for $x\text{La}_2\text{O}_3$ -CMASBZ ($x=0.0, 0.5, 1.0, 3.0$); (c) Inverse convolution fitted peaks in the range of 800~1200 cm^{-1} for $x\text{La}_2\text{O}_3$ -CMASBZ ($x=0.0, 0.5, 1.0, 3.0$); (d) area share of deconvolution cells.

3.3. Crystallization and thermal properties analysis

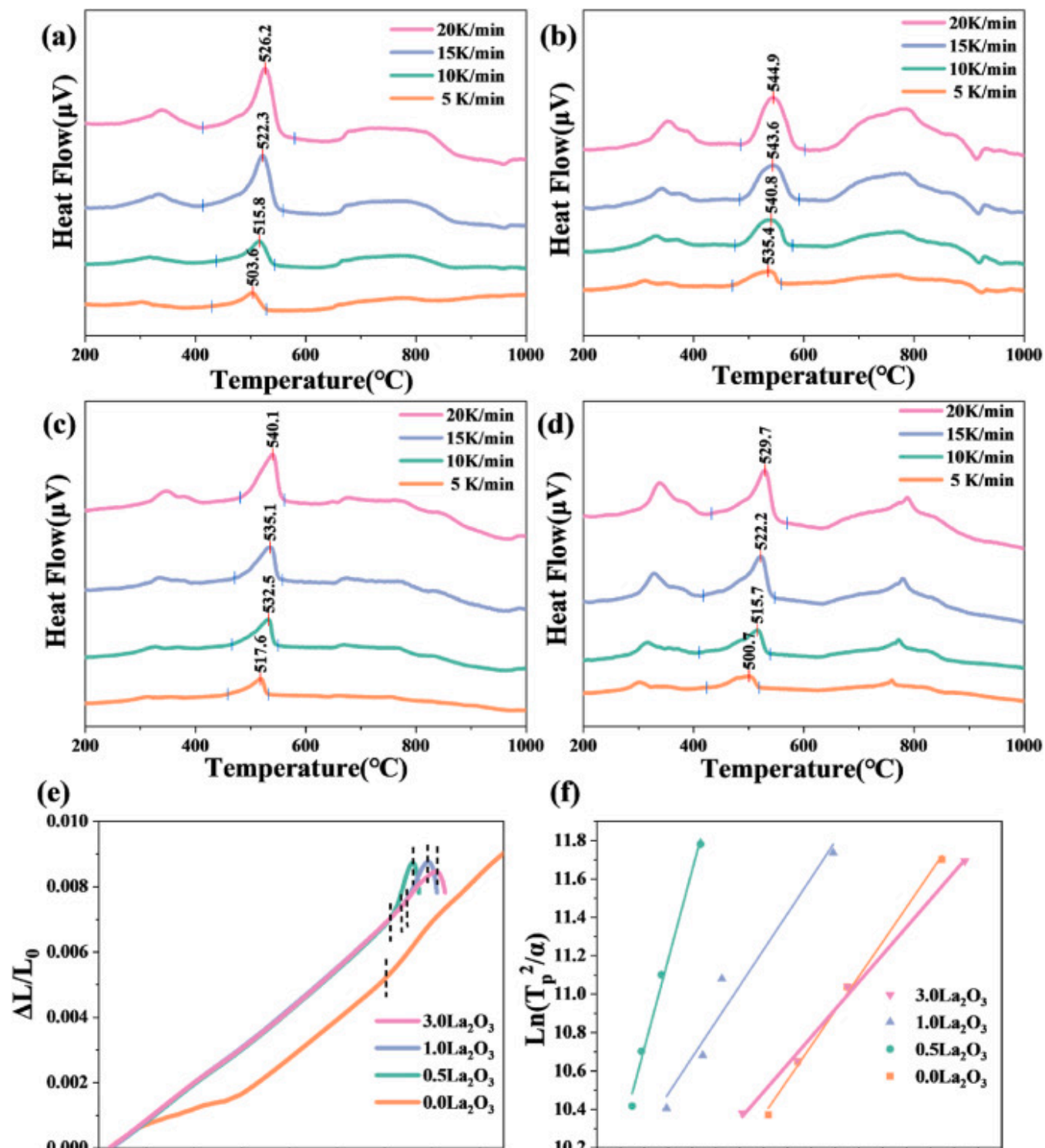
To investigate the crystallization behaviour of the xLa₂O₃-CMASBZ glass-ceramic system, the crystallization peak temperatures (T_p) were measured by differential scanning calorimetry (DSC) at heating rates of 5, 10, 15, and 20 Kmin⁻¹. The obtained DSC curves are shown in Fig. 4(a–d). The crystallization peak temperature increases with the rise of the heating rate, and the peak shape becomes sharper. During the crystallization process of glass, a relatively slow heating rate allows sufficient time for crystallization, thereby reducing the instantaneous crystallization rate. This results in a lower crystallization peak temperature and a more gradual exothermic peak. Conversely, a higher heating rate accelerates the crystallization process, leading to an elevated crystallization peak temperature and a sharper peak shape [46,47]. The Kissinger equation can be used to calculate the crystallization kinetics activation energy for the xLa₂O₃-CMASBZ system. The Kissinger equation is as follows [48,49]:

$$\ln \left(\frac{T_p^2}{\alpha} \right) = \frac{E_c}{RT_p} + K_1$$

Where T_p (K) is the crystallization peak temperature, α is the rate of temperature increase (K/min), E_c is the activation energy of crystallization, R is the gas constant 8.314 J/(mol·K), and K_1 is a constant. The relationship between $\ln \left(\frac{T_p^2}{\alpha} \right)$ and $1000/T_p$ from the experimental data is plotted using the linear-fitting method in Fig. 4(f). And the calculated R^2 are greater than 0.99, ensuring the accuracy of the experimental data. The Avrami parameter, n , which is related to the mechanism of the crystallization process, is determined from the crystallization exotherm using the equation given by Augis and Bennett [50].

$$n = \frac{2.5RT_p^2}{E_c \Delta T_{FWHM}}$$

Where ΔT_{FWHM} is the width of the crystallization peak at half maximum. The Avrami constant n close to 1, 2, 3, and 4 means one-dimensional growth, two-dimensional crystallization, three-dimensional growth (bulk crystallization), and homogeneous crystallization, respectively [51].





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Fig. 4. The DSC curves of glass-ceramic system at different heating rates (a) 0.0La₂O₃-CMASBZ; (b) 0.5La₂O₃-CMASBZ; (c) 1.0La₂O₃-CMASBZ; (d) 3.0La₂O₃-CMASBZ; (e) thermal expansion curve of xLa₂O₃-CMASBZ (x=0.0, 0.5, 1.0, 3.0) glass-ceramic; (f) fitting results with Kissinger model in the linear regression.

The effect of La₂O₃ content on the energy required for glass crystallization (E_c) varies from 299.73 to 767.64 kJ/mol and then decreases back to 237.95 kJ/mol. The incorporation of La₂O₃ induces changes in the glass crystal phase and results in a significant fluctuation in the crystallization activation energy. Specifically, the crystallization activation energy of MgSiO₃ is reported to be 718 kJ/mol [52], while that of diopside and monticellite ranges approximately between 322.68 and 477.10 kJ/mol [53]. Additionally, the Avrami index (n) increases from 1.94 in the 0La₂O₃-CMASBZ glass to 2.56 in the 3.0La₂O₃-CMASBZ glass. Initially, Ca_{0.88}Mg_{1.22}SiO₄ forms two-dimensional crystals. As the lanthanum oxide content increases, the crystal structures in the glass-ceramics evolve into one-dimensional configurations before reverting to two-dimensional arrangements.

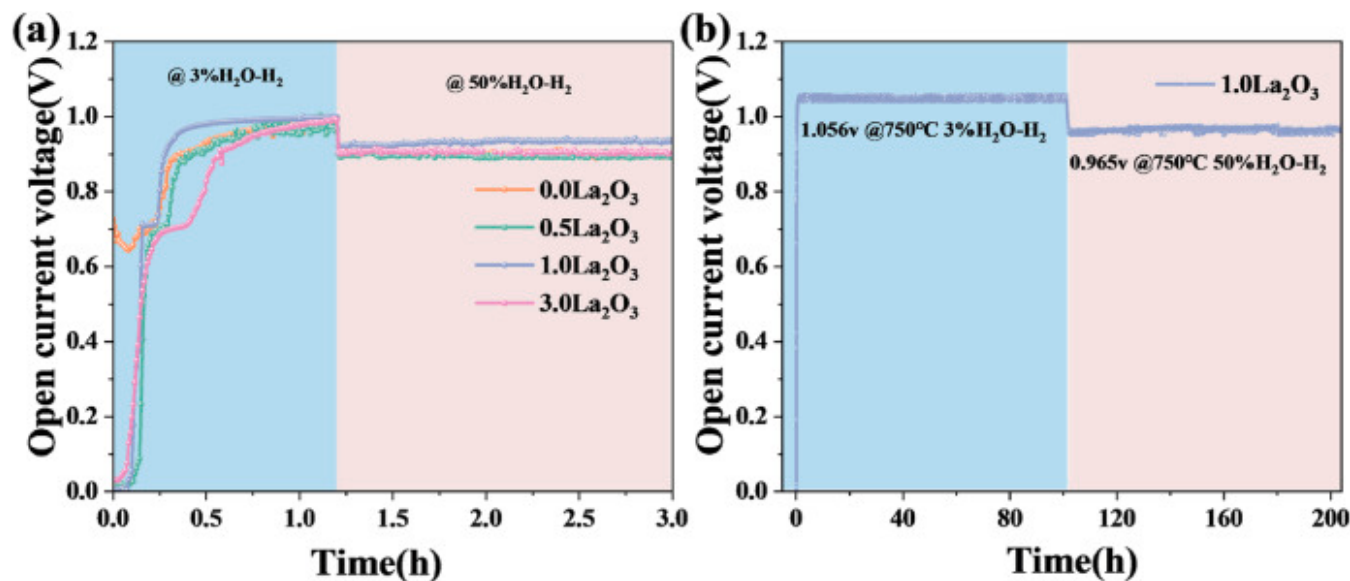
The CTE of the glass is a critical criterion for evaluating its applicability in SOECs stack sealing application. The CTE curves of xLa₂O₃-CMASBZ glass-ceramics from room temperature to 850°C are shown in Fig. 4(e). The related fitting values are listed in Table 1. The incorporation of La₂O₃ enhances the thermal expansion coefficient of glass-ceramic, and effectively suppresses the precipitation of the unfavorable crystalline phase MgO (CTE = $9.08 \times 10^{-6} \text{K}^{-1}$) [54]. The CTE values of the 1.0La₂O₃ and 3.0La₂O₃ glass-ceramics are $12.3 \times 10^{-6} \text{K}^{-1}$ and $12.2 \times 10^{-6} \text{K}^{-1}$, respectively, which matches perfectly with those of YSZ ($10.5 \times 10^{-6} \text{K}^{-1}$) and SUS430 ($12.5 \times 10^{-6} \text{K}^{-1}$) [55]. Consequently, CTE mismatch during the sealing process and stress concentration in the joint can be ignored. More importantly, given the phase stability of xLa₂O₃-CMASBZ at 850°C for 100h, the CTE of the sealant will remain relatively constant within the joint. Additionally, both the softening temperature (T_s) and the glass transition temperature (T_g) of the four samples decrease with the increasing La₂O₃ content.

Table 1. Summarization of the thermophysical properties of xLa₂O₃-CMASBZ (x=0,0.5,1.0,3.0).

Samples	CTE(*10 ⁻⁶ /K)	T _g (°C)	T _s (°C)	T _p (°C)	E _c (KJ/mol)	ΔT _{FWHM}	n
0.0La ₂ O ₃	9.58	654.4	803.4	503.6	299.73	18.0	1.94
0.5La ₂ O ₃	13.70	648.7	780.6	535.4	767.64	25.8	0.56
1.0La ₂ O ₃	12.30	640.6	768.5	517.6	331.57	16.15	2.14
3.0La ₂ O ₃	12.20	632.4	763.7	500.7	237.95	20.40	2.56

3.4. Sealing performance analysis

Fig. 5(a) demonstrates the sealing performance of xLa₂O₃-CMASBZ glass-ceramics through in-situ OCV characterization of NiO-YSZ/YSZ half-cells under reducing atmospheres (3%H₂O-H₂ and 50%H₂O-H₂). Based on the Nernst equation, the OCV is directly related to the partial pressure of the fuel and oxidant. OCV can serve as an indicator to evaluate changes in the partial pressures of fuel and oxidant [[56], [57], [58], [59], [60]]. Fig. 5(a) shows that OCV increases initially and then decreases from 0.0 La₂O₃ to 3.0 La₂O₃ with the increase of La₂O₃ substitution, reaching its maximum value at 1.0 La₂O₃. Specifically, the OCV values under 3%H₂O-H₂ conditions are 0.97, 0.98, 1.01, and 0.99V for 0.0, 0.5, 1.0, and 3.0% La₂O₃, respectively. In contrast, the values under 50%H₂O-H₂ conditions are 0.88V, 0.89V, 0.94V, and 0.90V, respectively. According to Tong et al. [61], the OCV of Bi₂O₃-SiO₂-BaO-R_xO_y (R = K, Zn, etc.) glass with added Al₂O₃ remains stable at approximately 1.0V at 750°C. Fig. 5(b) demonstrates that the voltage can be maintained steadily at 1.056V and 0.965V during a 100-h degradation test. This humidity-resistant sealing behavior confirms that 1.0 La₂O₃-CMASBZ is a highly promising candidate for SOCs applications.



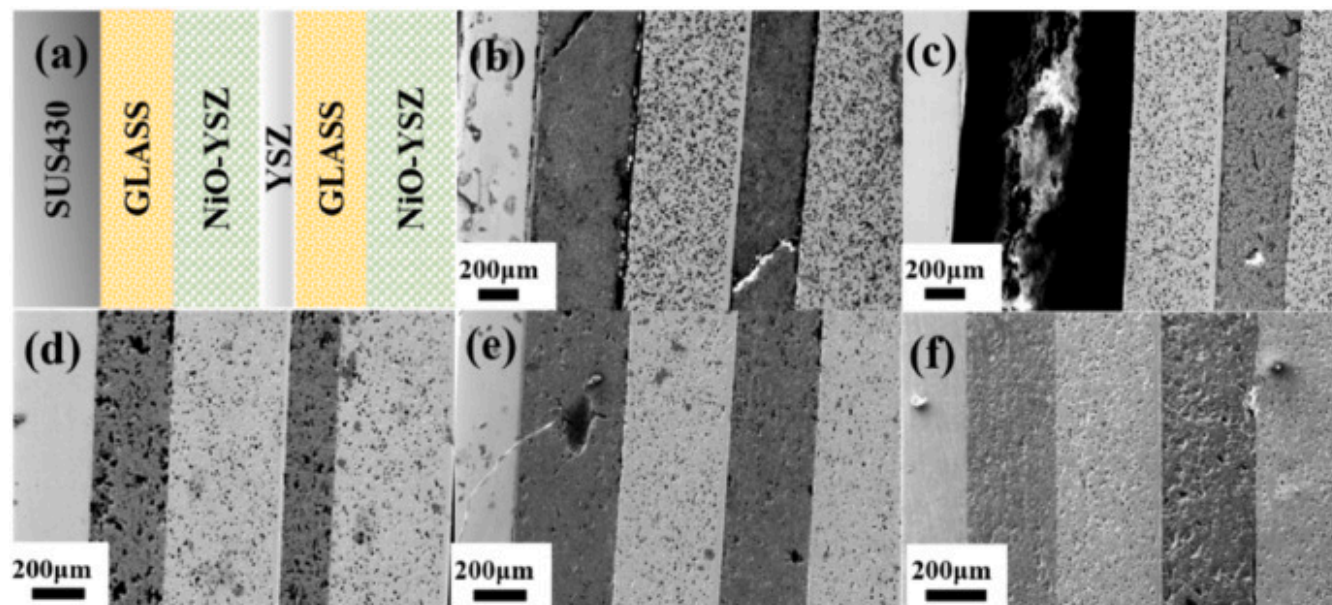
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Fig. 5. (a) The OCV of $x\text{La}_2\text{O}_3$ -CMASBZ ($x=0,0.5,1.0,3.0$) glass-ceramics at 850°C for $3\%\text{H}_2\text{O}-\text{H}_2$ and $50\%\text{H}_2\text{O}-\text{H}_2$; (b) The OCV of $1.0\text{La}_2\text{O}_3$ -CMASBZ glass-ceramic at 750°C for $3\%\text{H}_2\text{O}-\text{H}_2$ and $50\%\text{H}_2\text{O}-\text{H}_2$.

3.5. Cross-sectional SEM analysis

To investigate the bonding between CMASBZ-based glass-ceramics seal and SOECs adjacent joints, as well as to observe whether any chemical reactions occur, the cross-sectional morphology after heat treatment at 850°C for 5 h or 100 h in air is presented in Fig. 6. $0.0\text{La}_2\text{O}_3$ -CMASBZ exhibits pronounced interfacial cracks with YSZ electrolyte, NiO/YSZ anode, and SUS430 interconnect, accompanied by delamination during thermal cycling, which indicates insufficient interfacial compatibility. As shown in Fig. 6(d), there are numerous disconnected holes in the ceramic interface, which is attributed to the excessively high softening temperature of the $0.5\text{La}_2\text{O}_3$ glass-ceramic and its difficulty in crystallization. In contrast, glass ceramics exhibit better interface contact when the content of La_2O_3 exceeds 1%, suggesting the La_2O_3 addition enhances the interfacial wetting performance with YSZ electrolyte, NiO-YSZ electrode, and SUS430 interconnector. Based on these findings, the $1.0\text{La}_2\text{O}_3$ glass-ceramic sealing material was selected for further investigation into its long-term sealing performance under reducing and oxidizing conditions.

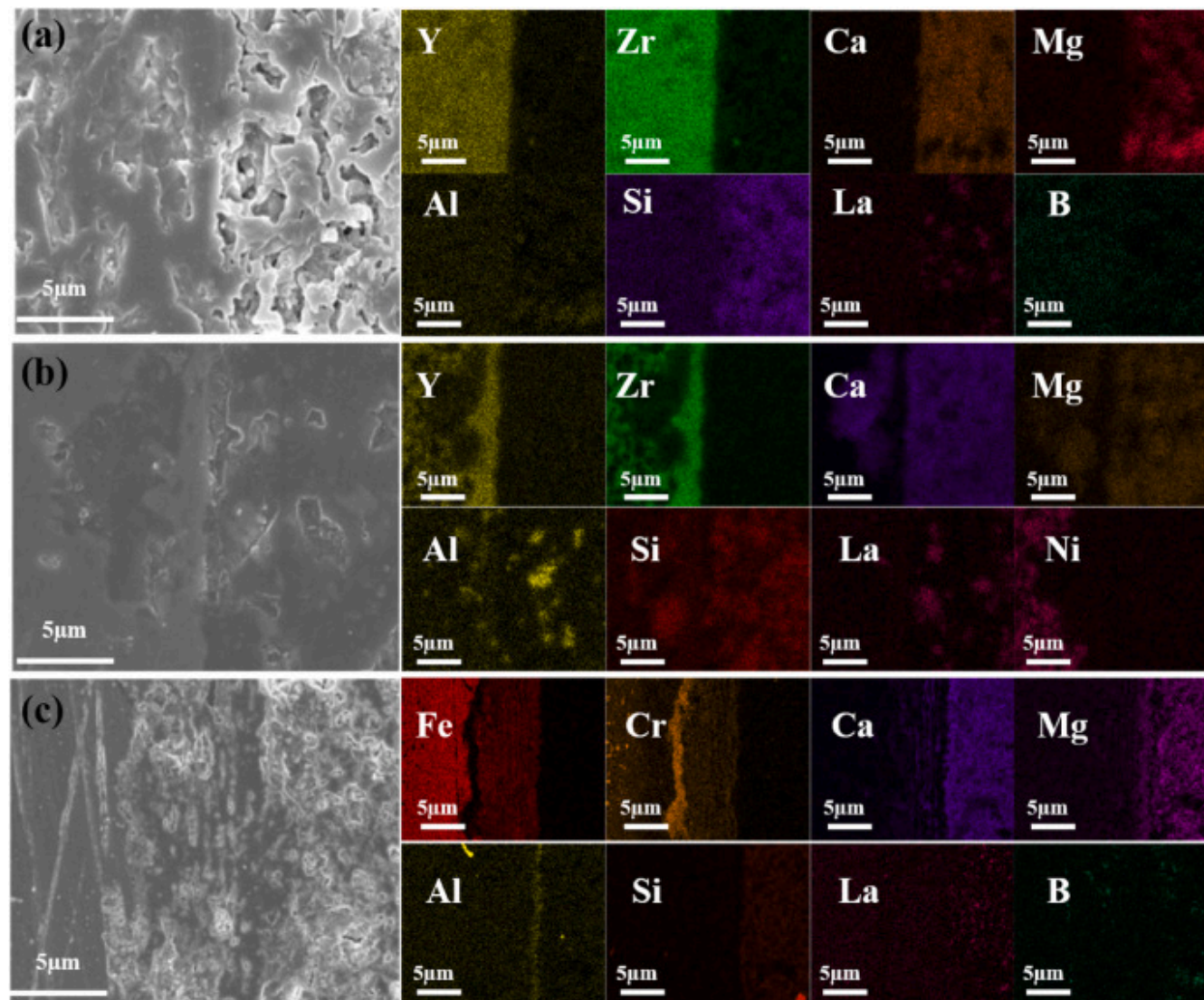


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Fig. 6. (a) SOECs component stacking diagram. The cross-sectional microstructure of 0.0La₂O₃-CMASBZ glass-ceramic system in air at (b) 850°C-5h and (c) 850°C-100h. The cross-sectional microstructure of glass-ceramic system at 850°C-5h in air for (d) 0.5 La₂O₃-CMASBZ; (e) 1.0 La₂O₃-CMASBZ; (f) 3.0 La₂O₃-CMASBZ.

Fig. 7 presents high-magnification SEM analysis of the triple-phase interfaces in 1.0 La₂O₃-CMASBZ after exposure to air at 850°C for 100h. The interfacial microstructure demonstrates excellent contact integrity among all components. In Fig. 7(a), the 1.0 La₂O₃-CMASBZ/YSZ interface displays sharp elemental demarcation with minimal elemental migration, as confirmed by EDS mapping. This stability persists despite the formation of a porous structure due to NiO reduction during operation, which typically facilitates sealant infiltration. Notably, EDS mapping in Fig. 7(b) reveals selective elemental penetration: Ca, Mg, and Si migrate into pores, forming thermodynamically stable pyroxene phases. In contrast, Al and La exhibit interfacial segregation; the relatively large molecular size of La hinders its diffusion, leading to enrichment at the interface [21]. Fig. 7(c) illustrates a Cr₂O₃ barrier layer at the SUS430 interface, which effectively prevents 1.0 La₂O₃-CMASBZ diffusion into the metallic substrate. Additionally, Al₂O₃ plays a role in inhibiting Cr element diffusion.

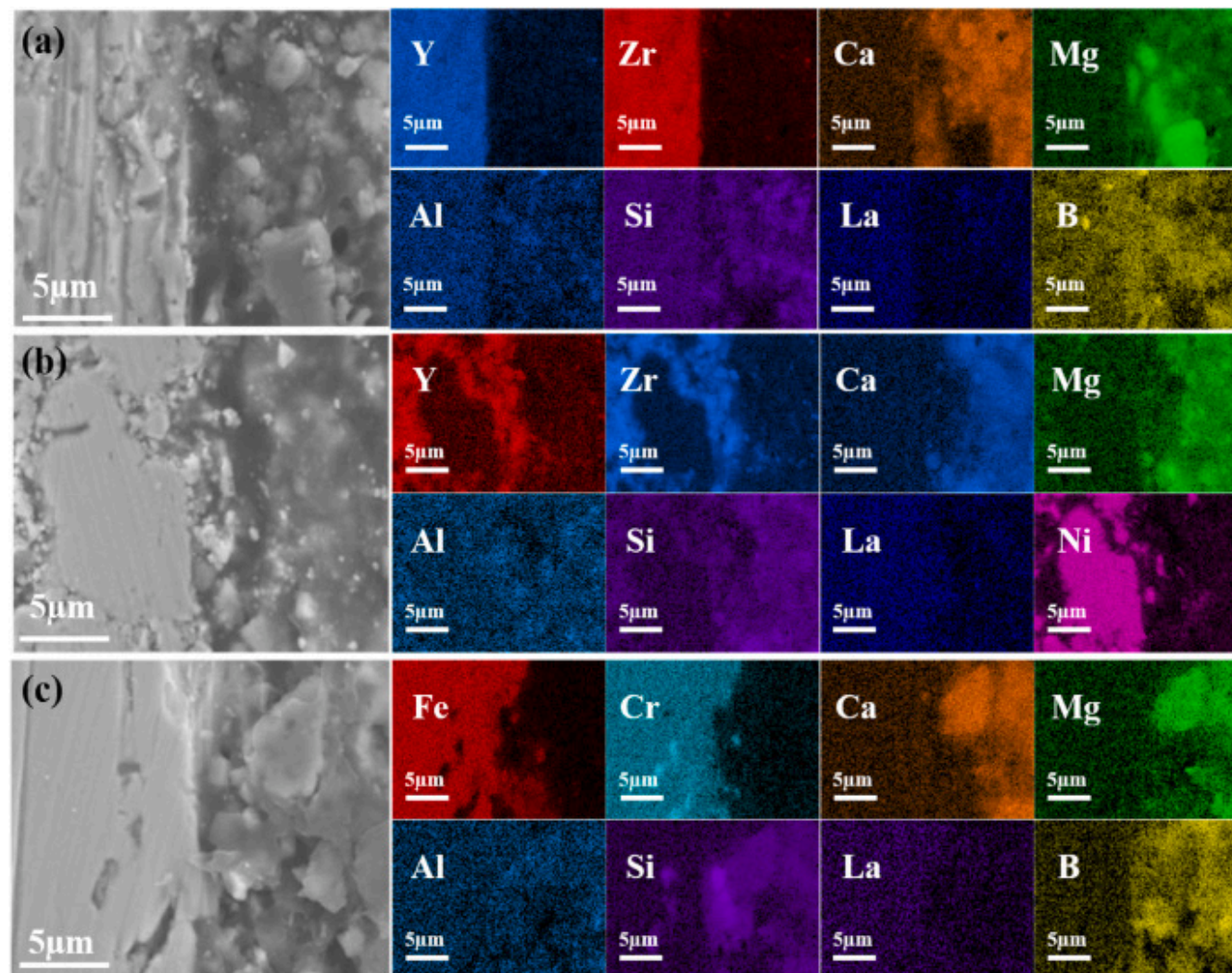


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Fig. 7. The micromorphology and EDS of 1.0 La₂O₃-CMASBZ with (a)YSZ; (b)NiO-YSZ; (c) SUS430 after heating treatment at 850°C –100h.

Fig. 8(a–c) displays SEM/EDS analysis of the 1.0 La₂O₃-CMASBZ/YSZ, 1.0 La₂O₃-CMASBZ/NiO-YSZ and 1.0 La₂O₃-CMASBZ/SUS430 interface following 100-h exposure to reducing condition (50% H₂O-(10% H₂-Ar)) at 850°C. The interfaces exhibit excellent adhesion, remaining free from defects such as cracks, bubbles, or leakage channels. EDS mapping reveals significant interfacial diffusion of Al, Ca, Mg, Si, and B as depicted in Fig. 8(a–c). This phenomenon can be attributed to water vapor-enhanced cation diffusion. However, such diffusion exhibits negligible impact on the sealing integrity of the 1.0La₂O₃-CMASBZ composite. The 1.0 La₂O₃-CMASBZ seal demonstrates robust interfacial stability and chemical compatibility under prolonged high-temperature operation in both wet reducing (50% H₂O-(10% H₂-Ar)) and oxidizing environments.



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Fig. 8. The micromorphology and EDS of 1.0 La₂O₃- CMASBZ with (a)YSZ; (b)NiO/YSZ; (c) SUS430 after heating treatment at 850°C – 100h in 50% H₂O-(10% H₂-Ar) atmospheres.

4. Conclusions

In summary, this study systematically investigates the role of La₂O₃ in tailoring the thermophysical properties, crystallization kinetics, and interfacial stability of CMASBZL glass-ceramics for solid oxide electrolytic cell (SOECs). Raman spectroscopy reveals that La₂O₃ addition induces structural depolymerization through conversion of bridging oxygen to non-bridging oxygen species, and transformation of Q² to Q⁰ coordination states in [SiO₄] tetrahedra. These modifications enhance network flexibility while maintaining thermal stability. In addition, the La₂O₃ content significantly modulates thermal behavior in a non-monotonic pattern: The CTE first increases from $9.85 \times 10^{-6} \text{K}^{-1}$ to $13.7 \times 10^{-6} \text{K}^{-1}$ before stabilizing at $12.2 \times 10^{-6} \text{K}^{-1}$, concurrently with reduction in glass transition from 654.4 to 632.4 °C and softening temperatures from 803.4 to 763.7 °C. This controlled CTE adjustment enables optimal thermal matching with SOECs stack components. The 1.0 La₂O₃-CMASBZ demonstrates exceptional operational reliability. The open-circuit voltages of 1.056V (3% H₂O-H₂) and 0.956V (50% H₂O-H₂) are stable without detectable degradation under SOCs conditions. Cross-sectional analysis confirmed chemical compatibility across YSZ, NiO/YSZ, and SUS430 interfaces, with effective suppression of detrimental cation interdiffusion - particularly Cr³⁺ and Ni²⁺ - under both reducing (50% H₂O-(10% H₂-Ar)) and oxidizing atmospheres. These results demonstrate that 1.0 La₂O₃-CMASBZ is a high-performance sealing solution for next-generation SOECs systems.

CRedit authorship contribution statement

Zihao Xu: Writing – original draft, Investigation, Data curation, Conceptualization. **Fushun Zhang:** Writing – review & editing, Supervision, Conceptualization. **Xu Dang:** Visualization, Methodology, Investigation. **Chaoyi Wang:** Visualization, Methodology, Investigation. **Xiaobo He:** Writing – original draft. **Hongtao Chang:** Funding acquisition. **Jun Peng:** Funding acquisition. **Liuzhen Bian:** Writing – review & editing, Supervision, Conceptualization. **Yunting Hou:** Writing – review & editing. **Shengli An:** Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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